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Nationality - Canadian
Date of Birth - 20.11.1995
Sex - Male

Evert Nasedkin, PhD

Curriculum Vitae

About me

Through my training as an astronomer I have gained extensive experience in scientific data analysis, modelling of physical processes, open-source software development, and in the communication of complex results to a variety of audiences.

Professional Experience

Current **Postdoctoral Fellow**, *Trinity College Dublin*, Dublin, IR.
- Implemented open-source, GPU accelerated framework for performing inference of time-resolved atmospheric properties of exoplanets.
- Awarded 8.7 hours observing time with NASA's James Webb Space Telescope.
- Awarded 6.4M CPU-hours of compute for atmospheric analyses, Meluxina Class B.
- Awarded 2500 node-hours of compute (CPU+GPU) for testing, Meluxina Class C.
- Co-authored multiple high-impact publications, presented findings at premier international conferences, and organised media releases for key results.
- Mentored two bachelor's student projects in modelling and analysis of exoplanet spectra.
- Organised multiple workshops, conference sessions, and outreach events.

2020-2024 **IMPRS PhD Fellow**, *Max Planck Institute for Astronomy*, Heidelberg, DE.
- Pioneered a novel Bayesian atmospheric retrieval framework within the open-source petitRADTRANS package, fitting atmospheric models to spectroscopic data.
- Analysed hyperspectral imaging datasets using python-based image processing tools.
- Published 3 first-author papers, presented results at 11 international conferences.
- Organised multiple workshops and seminars with an international audience.
- Served as PhD student representative, led efforts to improve onboarding processes.
- Mentored a 6 month bachelor's student project.

09-12 2019 **Semester Project**, *Institute for Astronomy*, ETH Zürich, Zürich, CH.
- Processed and visualised 3D hydrodynamic simulations of planet formation.

05-08 2017 **Research Assistant**, *Institute for Astronomy*, ETH Zürich, Zürich, CH.
- Investigated mechanical properties at cryogenic conditions for astronomical instrument.

08-12 2016 **Research Assistant**, *nEXO Collaboration*, McGill University, Montreal, CA.
- Simulated and assembled test source for the nEXO neutrino experiment.

2015-2016 **Research Assistant**, *DEAP-3600 Dark Matter Search*, SNOLAB, Sudbury, CA.
- Implemented and automated analyses for characterising detector behaviour.

2014-2016 **Aerodynamics Team Member**, *FSAE Student Design Team*, Waterloo, CA.
Performed computational fluid dynamics experiments; assisted with manufacturing.

Education

2020-2024 **PhD, Astronomy**, University of Heidelberg, Heidelberg, DE,
Max Planck Institute for Astronomy, Heidelberg, DE.
Thesis: "Atmospheric Characterisation of Directly Imaged Exoplanets".
Grade: Magna cum laude.

2018–2020 **Master of Science, Physics**, ETH Zürich, Zürich, CH.

2013–2018 **Bachelor of Science, Physics**, University of Waterloo, Waterloo, CA.

Computing & Technical Skills

Projects

petitRADTRANS Codeveloper of widely used pRT package (>300 citations). Wrote module to perform atmospheric retrievals on exoplanets and brown dwarfs. `https://gitlab.com/mauricemolli/petitRADTRANS`

Data reduction for IFUs Developed pipelines to streamline the post-processing of high-contrast hyperspectral data using state-of-the-art algorithms.

PACO Contributed a python implementation of the PAtch COvariance (PACO) PSF subtraction algorithm to the open-source `VIP-HCI` package.

Technologies

Python Expert – Open source design and development, extensive use of data analysis and machine learning packages (`numpy`, `JAX`, `pytorch`, `scikit-learn`, `pandas`).

Fortran Intermediate – Contributed to pRT radiative transfer backend.

C++ Intermediate – Automated analysis for DEAP-3600 detector characterisation.

HPC Awarded class B and C programs on EuroHPC cluster Meluxina.

Tools Use of `git` for version control, automated testing; package release via PyPI; experience with Linux, MacOS, Windows, bash scripting, and command line interfaces.

AI Careful use of Copilot, LLMs (e.g. Gemini) and agentic tools to optimize workflows.

Mentorship, Teaching and Supervision

09-11 2025 **Supervision of fourth year capstone project**, *Trinity College Dublin*, Dublin, IE.

05-07 2025 **Supervision of SURE Summer intern**, *Trinity College Dublin*, Dublin, IE.

01-04 2025 **Computational Physics Lab Instructor**, *Trinity College Dublin*, Dublin, IE.

04-12 2022 **Supervision of Bachelor's Miniprojekt**, *MPIA*, Heidelberg, DE.

2022-2023 **Astro-lab Tutor**, *University of Heidelberg*, Heidelberg, DE.

2017-2018 **Teaching Assistant**, *PHYS 111*, Waterloo, CA.

01-04 2015 **English Language Teacher**, *TOBB ETU University*, Ankara, TR.

Languages

English: Native. **Spanish**: Conversational. **German**: Basic. **French**: Basic.

Selected Publications

1 **Nasedkin**, Schrader, Vos et al. "The JWST weather report: temperature variations, auroral heating and static clouds on SIMP-0136", A&A, 687, A298 (2025).

2 Hoch, Rowland, Petrus, **Nasedkin**, et al. "Silicate clouds and circumplanetary disk in the YSES-1 System", Nature 643, 8073 (2025).

3 **Nasedkin**, Mollière, Lacour et al. "Four-of-a-kind? Comprehensive atmospheric characterisation of the HR 8799 planets with VLTI/GRAVITY", A&A, 687, A298 (2024).

4 **Nasedkin**, Mollière, and Blain "Atmospheric retrievals with petitRADTRANS", JOSS 9, 6, 5875 (2024).

5 **Nasedkin**, Mollière, Wang et al. "Impacts of high-contrast image processing on atmospheric retrievals", A&A, 678, A41 (2023).

6 Vasist, Rozet, Absil et al. (incl. **Nasedkin**) "Neural posterior estimation for exoplanetary atmospheric retrieval", A&A, 672, A147 (2023).

7 Christiaens, Gonzalez, Farkas et al. (incl. **Nasedkin**) "VIP: A Python package for high-contrast imaging", JOSS, 8, 4774 (2023).

8 Patapis, **Nasedkin**, Cugno et al. "Direct emission spectroscopy of exoplanets with the medium resolution imaging spectrometer on board JWST MIRI. I. Molecular mapping and sensitivity to instrumental effects", A&A, 658, A72 (2022).